

Announcements

- CSTEM Symposium, TOMORROW!
 - 9am - 5pm
 - Athletics Building
 - Submit a selfe & 1 paragraph reflection for extra points on Final Exam for Symposium attendance.
 - Banquet (Invitation Only) SAC Ballroom A
5:30 - 7:30 pm
 - Milestone 2 & 3 to be posted soon
 - April 12 Milestone 2
 - May 3 Milestone 3
-

Question 1 Difficulty: 1

An if statement must always be followed by an else.

True 2 (13.33 %)
⇒ False 13 (86.67 %)

Average Grade: 0.87 / 1 (86.67 %)
Standard Deviation: 33.99 %
Point Biserial: 0.48
Discrimination Index: 25.00 %

Question 2 Difficulty: 1

What is the output of the following code?

```
$val = 2;  
if($val == 2){  
    echo "delay";  
}else if($val == 2){  
    echo "share";  
}else{  
    echo "judge";  
}
```

Handwritten notes:
- "2" is a string, not an integer.
- "==" is for same value & data type.
- "==" is for same value.

delay 1 (6.67 %)
⇒ share 13 (86.67 %)
judge 1 (6.67 %)
Nothing, this is an error. 0 (0 %)

Average Grade: 0.87 / 1 (86.67 %)
Standard Deviation: 33.99 %
Point Biserial: 0.48
Discrimination Index: 25.00 %

Question 3 Difficulty: 1

Write a while loop that prints the numbers from 7 to 2.

Expand Responses

```
$num = 7;  
while ($num <= 2){  
    echo $num;  
    $num --;  
}
```

Average Grade: 0 / 3 (0 %)
Standard Deviation: 0.00 %
Point Biserial: n/a
Discrimination Index: 0.00 %

Question 4 Difficulty: 1

What is the output of the following code?

```
$numbers = [1, 2, 3];  
foreach ($numbers as $num) {  
    echo $num * 2 . " ";  
}
```

1 2 3 0 (0 %)
⇒ 2 4 6 15 (100 %)
Nothing 0 (0 %)
0 1 2 0 (0 %)

Average Grade: 1 / 1 (100 %)
Standard Deviation: 0.00 %
Point Biserial: n/a
Discrimination Index: 0.00 %

Question 5 Difficulty: 1

In class, we discussed three ways to control when the PHP interpreter will execute statements. Two of those ways are ____ and ____.

Answers

selection 8 (53.33 %)
sequence 4 (26.67 %)
(?)^(repetition)|(iteration) 9 (60 %)
Other 9

Average Grade: 0.93 / 2 (46.66 %)
Standard Deviation: 29.31 %
Point Biserial: n/a
Discrimination Index: 66.65 %

selection
sequence

selection
sequence
repetition / iteration

Functions

```
function apply_discount ($price=10, $discount=0.10) {  
    $discount_amt = $price * $discount;  
    return $price - $discount_amt;  
}
```

parameters

default value

$\$price = \text{apply_discount}()$; — uses default values for both params
 $\$price = \text{apply_discount}(2.5)$; — uses 2.5 for price & default for discount
 $\$price = \text{apply_discount}(\text{discount}=.25)$; — uses default for price & .25 for discount
 $\$price = \text{apply_discount}(20, 0.2)$; — uses the given arguments for both.

parameters are the variables in the function definition
arguments are the values (literal or variables) passed to the function when called.

Functions & Arrays

Arrays can be used as params or returned from a function

```
$rates = ['uk' => 0.81, 'eu' => 0.93, 'jp' => 113.21];  
  
function calculate_global_prices($usd, $exchange_rates) {  
    $prices = [  
        'pound' => $usd * $exchange_rates['uk'],  
        'euro'   => $usd * $exchange_rates['eu'],  
        'yen'    => $usd * $exchange_rates['jp']  
    ];  
}
```

associative array param

```

    'yen' => $usd * $exchange_rates['jp']
  ];
  return $prices;
}

```

returned array

```
$global_prices = calculate_global_prices(10.0, $rates);
```

Variadic Functions

Can give parameters as variable-length argument lists.
can have any # of arguments ≥ 1

```

function sum(...$numbers){
  $total = 0; — variable for total

  foreach($numbers as $num){ — loop through
    $total += $num; — add current value to total
  }

  return $total; — return final total when done.
}

```

```

function family_print($last, ...$first){
  $txt = '';

  foreach($first as $name){
    $txt .= "Hi, $name $last!<br>";
  }

  return $txt;
}

```

Built-In Functions

include 7 place existing file inside another file

using the functions

include } place existing file inside another file
require }

include - try to render the page & give warning if error
require - won't render page if error.